

```
> restart
```

Guessing examples with the package

DalgGuessing from

<https://github.com/T3gu1a/D-algebraic-Guessing>

```
>
>
> with(DalgGuessing)
                                [DalgFunGuess, modDalgFunGuess] (1)
> with(CodeTools, CPUTime) #CPUTime is often used to display the CPU time of computations
                                [CPUTime] (2)
>
```

▼ DalgFunGuess

```
>
```

▼ **Example 1:** $\zeta(2n+2), n=0, 1, 2 \dots$

```
>
> L := [seq(Zeta(2*j+2), j=0..12)] #13 terms
L := [  $\frac{\pi^2}{6}, \frac{\pi^4}{90}, \frac{\pi^6}{945}, \frac{\pi^8}{9450}, \frac{\pi^{10}}{93555}, \frac{691 \pi^{12}}{638512875}, \frac{2 \pi^{14}}{18243225}, \frac{3617 \pi^{16}}{325641566250},$  (1.1.1)
 $\frac{43867 \pi^{18}}{38979295480125}, \frac{174611 \pi^{20}}{1531329465290625}, \frac{155366 \pi^{22}}{13447856940643125},$ 
 $\frac{236364091 \pi^{24}}{201919571963756521875}, \frac{1315862 \pi^{26}}{11094481976030578125} ]$ 
```

```
> DalgFunGuess(L, degADE=2, degPoly=1, startfromord=2)

$$-C y(x)^2 + \frac{5_C \left( \frac{d}{dx} y(x) \right)}{2} - 2_C x \left( \frac{d}{dx} y(x) \right) y(x) + C x \left( \frac{d^2}{dx^2} y(x) \right) = 0 \quad (1.1.2)$$

```

```
> DalgFunGuess(L[1..9], degADE=2, degPoly=1, startfromord=2, allPolyDeg=true)
#9 terms

$$-C y(x)^2 + \frac{5_C \left( \frac{d}{dx} y(x) \right)}{2} - 2_C x \left( \frac{d}{dx} y(x) \right) y(x) + C x \left( \frac{d^2}{dx^2} y(x) \right) = 0 \quad (1.1.3)$$

```

```
> CPUTime(DalgFunGuess(L[1..7], degADE=2, degPoly=1, startfromord=2,
```

$allPolyDeg = true, sparsity = \frac{1}{2} \Big) \Big) \#7 \text{ terms}$

$$0.563, -\frac{2_C y(x)^2}{5} + _C \left(\frac{d}{dx} y(x) \right) - \frac{4_C x \left(\frac{d}{dx} y(x) \right) y(x)}{5} + \frac{2_C x \left(\frac{d^2}{dx^2} y(x) \right)}{5} = 0 \quad (1.1.4)$$

Example 2: Gregory coefficients $\frac{x}{\ln(1+x)} = 1 + \sum_{n=0}^{\infty} G_n x^n$

```
> f := (x/log(1+x) - 1)/x :
> L := PolynomialTools:-CoefficientList(convert(series(f, x, 21), polynom), x)
L := [ 1/2, -1/12, 1/24, -19/720, 3/160, -863/60480, 275/24192, -33953/3628800, 8183/1036800,
-3250433/479001600, 4671/788480, -13695779093/2615348736000, 2224234463/475517952000,
-132282840127/31384184832000, 2639651053/689762304000, -111956703448001/32011868528640000, 50188465/15613165568,
-2334028946344463/7860144949376000, 301124035185049/109285437800448000 ]
```

```
> numelems(L)
19
```

(1.2.2)

```
> DalgFunGuess(L, degADE=2, degPoly=2, startfromord=2) #19 terms
\_C y(x)^2 + 2\_C x \left( \frac{d}{dx} y(x) \right) y(x) + (2\_C x + 3\_C) \left( \frac{d}{dx} y(x) \right) + (\_C x^2
+ \_C x) \left( \frac{d^2}{dx^2} y(x) \right) = 0
```

(1.2.3)

```
> CPUTime(DalgFunGuess(L[1..11], degADE=2, degPoly=2, startfromord=2,
allPolyDeg=true)) #11 terms
0.375, \_C y(x)^2 + 2\_C x \left( \frac{d}{dx} y(x) \right) y(x) + (2\_C x + 3\_C) \left( \frac{d}{dx} y(x) \right) + (\_C x^2
+ \_C x) \left( \frac{d^2}{dx^2} y(x) \right) = 0
```

(1.2.4)

```
> CPUTime(DalgFunGuess(L[1..9], degADE=2, degPoly=2, startfromord=2,
```

$$\begin{aligned}
 & allPolyDeg = true, sparsity = \frac{1}{2} \Big) \Big) \#9 \text{ terms and } \textbf{wrong} \text{ guess} \\
 13.625, & \left(-\frac{136876076071244256}{149231401475539} _C x - \frac{57404081005704237}{149231401475539} _C x^2 \right) \left(\frac{d}{dx} \right. \\
 & y(x) \Big) y(x) + \left(\frac{568456138313805960}{149231401475539} _C x \right. \\
 & + \frac{521098571596151412}{149231401475539} _C x^2 \Big) \left(\frac{d}{dx} y(x) \right)^2 + _C x^2 y(x)^2 \\
 & + \left(\frac{115808859621663588}{149231401475539} _C x + \frac{16903414409689274}{149231401475539} _C x^2 \right) \left(\frac{d}{dx} y(x) \right) \\
 & + \left(-\frac{523273442370}{149231401475539} _C x + \frac{6212024644107945}{298462802951078} _C x^2 \right) \left(\frac{d^2}{dx^2} y(x) \right) = 0
 \end{aligned} \tag{1.2.5}$$

> CPUTime(DalgFunGuess(L[1..10], degADE = 2, degPoly = 2, startfromord = 2,

$$\begin{aligned}
 & allPolyDeg = true, sparsity = \frac{1}{2} \Big) \Big) \#10 \text{ terms} \\
 1.063, & _C y(x)^2 + 2 _C x \left(\frac{d}{dx} y(x) \right) y(x) + (2 _C x + 3 _C) \left(\frac{d}{dx} y(x) \right) + (_C x^2 \\
 & + _C x) \left(\frac{d^2}{dx^2} y(x) \right) = 0
 \end{aligned} \tag{1.2.6}$$

>

Example 3: Dividing D-algebraic functions:

An instance of finding an algebraic differential equation satisfied by

ALL $\frac{f}{g}$, where $p(f) = q(g) = 0$

>

```

1  #The differential polynomials
2  p:=diff(y(x), x) - x*y(x)^2:
3  q:=-diff(z(x), x) + z(x) + x + 1:
4  sys := {p=0,q=0}: #Note: it is better to treat both ADEs separately as they are independent
5
6  #symbolic initial conditions
7  InitialConds:= {y(0) = theta, z(0) = beta}:
8
9  #Order is a global variable for truncating series solutions
10 Order :=19:
11
12 #Computing truncations of f and g of order Order
13 sol := dsolve(sys union InitialConds {y(x), z(x)}, series):
14
15 # Extract the truncations
16 f_series := eval(y(x), sol):
17 g_series := eval(z(x), sol):
18
19 # Convert to polynomial
20 h_series := series(f_series/g_series, x, Order):
21 h_poly := convert(h_series, polynom):
22
23 # List of coefficients
24 L:=PolynomialTools-CoefficientListh_poly,x):

```

> simplify(L[1..5]) #5 first terms

$$\left[\frac{\theta}{\beta}, -\frac{\theta(1+\beta)}{\beta^2}, \frac{(2+(\theta+1)\beta^2+2\beta)\theta}{2\beta^3}, \right. \\ \left. -\frac{\left(2+\left(\theta+\frac{1}{3}\right)\beta^3+\left(\theta+\frac{2}{3}\right)\beta^2+2\beta\right)\theta}{2\beta^4}, \right. \\ \left. \frac{\left(4+\left(\theta^2+\theta+\frac{1}{6}\right)\beta^4+\left(2\theta-\frac{1}{3}\right)\beta^3+\left(2\theta+\frac{2}{3}\right)\beta^2+4\beta\right)\theta}{4\beta^5} \right] \quad (1.3.1)$$

> numelems(L) 19 (1.3.2)

> t, ADE := CPUTime(DalgFunGuess(L, degADE=2, degPoly=5, startfromord=1, inputConstants={beta, theta}))#19 terms

$$t, ADE := 1.672, \left(-\frac{2_C}{\theta^2} - \frac{2_Cx}{\theta^2} + \frac{2_Cx^2}{\theta} + \frac{2_Cx^3}{\theta} - \frac{Cx^4}{2} - \frac{Cx^5}{2} \right) y(x)^2 + \left(-\frac{2_C}{\theta} + 2_Cx + _Cx^2 \right) y(x) + \left(-\frac{2_C}{\theta} + _Cx^2 \right) \left(\frac{d}{dx} y(x) \right) = 0 \quad (1.3.3)$$

> CPUTime(DalgFunGuess(L[1..13], degADE=2, degPoly=5, startfromord=1, inputConstants={beta, theta}, allPolyDeg=true))#13 terms

$$6.250, \left(-\frac{2_C}{\theta^2} - \frac{2_Cx}{\theta^2} + \frac{2_Cx^2}{\theta} + \frac{2_Cx^3}{\theta} - \frac{Cx^4}{2} - \frac{Cx^5}{2} \right) y(x)^2 + \left(-\frac{2_C}{\theta} + 2_Cx + _Cx^2 \right) y(x) + \left(-\frac{2_C}{\theta} + _Cx^2 \right) \left(\frac{d}{dx} y(x) \right) = 0 \quad (1.3.4)$$

> CPUTime(DalgFunGuess(L[1..12], degADE=2, degPoly=5, startfromord=1, inputConstants={beta, theta}, allPolyDeg=true, sparsity=1/3, maxIteration=50))

#12 terms, would only succeed with luck

12.359, FAIL (1.3.5)

Eliminating θ

> $\rho_1 := \text{subs}([_C=2\cdot\theta^2, y(x)=y, \text{diff}(y(x), x)=y1], \text{lhs}(ADE))$

$$\rho_1 := (-\theta^2 x^5 - \theta^2 x^4 + 4\theta x^3 + 4\theta x^2 - 4x - 4) y^2 + (2\theta^2 x^2 + 4\theta^2 x - 4\theta) y + (2\theta^2 x^2 - 4\theta) y1 \quad (1.3.6)$$

> $\rho_2 := \text{subs}([_C=2\cdot\theta^2, y(x)=y, \text{diff}(y(x), x)=y1, \text{diff}(y(x), x, x)=y2],$

$$\begin{aligned} & \text{diff}(lhs(ADE), x) \\ \rho_2 := & (-5 \theta^2 x^4 - 4 \theta^2 x^3 + 12 \theta x^2 + 8 \theta x - 4) y^2 + 2 (-\theta^2 x^5 - \theta^2 x^4 + 4 \theta x^3 \\ & + 4 \theta x^2 - 4 x - 4) y y l + (4 \theta^2 x + 4 \theta^2) y + (2 \theta^2 x^2 + 4 \theta^2 x - 4 \theta) y l \\ & + 4 \theta^2 x y l + (2 \theta^2 x^2 - 4 \theta) y^2 \end{aligned} \quad (1.3.7)$$

$$\begin{aligned} & \text{rho} := \text{resultant}(\rho_1, \rho_2, \text{theta}) \\ \rho := & -1024 x^6 y^7 - 3072 x^5 y^7 - 3072 x^4 y^7 - 256 x^4 y^6 - 1536 x^4 y^5 y l + 1024 x^4 y^5 y^2 \\ & - 2048 x^4 y^4 y l^2 - 1024 x^3 y^7 - 1536 x^3 y^6 - 4096 x^3 y^5 y l + 2048 x^3 y^5 y^2 \\ & - 4096 x^3 y^4 y l^2 - 1280 x^2 y^6 - 3072 x^2 y^5 y l + 1024 x^2 y^5 y^2 - 2048 x^2 y^4 y l^2 \\ & - 256 x^2 y^4 y l + 256 x^2 y^4 y^2 - 1024 x^2 y^3 y l^2 + 768 x^2 y^3 y l y^2 - 256 x^2 y^3 y^2 \\ & - 1536 x^2 y^2 y l^3 + 1024 x^2 y^2 y l^2 y^2 - 1024 x^2 y y l^4 - 512 x y^5 y l - 256 x y^5 \\ & - 1280 x y^4 y l + 768 x y^4 y^2 - 2816 x y^3 y l^2 + 1280 x y^3 y l y^2 - 256 x y^3 y^2 \\ & - 2560 x y^2 y l^3 + 1024 x y^2 y l^2 y^2 - 1024 x y y l^4 + 256 y^6 - 256 y^5 - 768 y^4 y l \\ & + 256 y^4 y^2 - 1024 y^3 y l^2 + 256 y^3 y l y^2 - 512 y^2 y l^3 \end{aligned} \quad (1.3.8)$$

$$\begin{aligned} & \text{factor}(\text{rho}) \\ -256 y (& 4 x^6 y^6 + 12 x^5 y^6 + 12 x^4 y^6 + x^4 y^5 + 6 x^4 y^4 y l - 4 x^4 y^4 y^2 + 8 x^4 y^3 y l^2 \\ & + 4 x^3 y^6 + 6 x^3 y^5 + 16 x^3 y^4 y l - 8 x^3 y^4 y^2 + 16 x^3 y^3 y l^2 + 5 x^2 y^5 + 12 x^2 y^4 y l \\ & - 4 x^2 y^4 y^2 + 8 x^2 y^3 y l^2 + x^2 y^3 y l - x^2 y^3 y^2 + 4 x^2 y^2 y l^2 - 3 x^2 y^2 y l y^2 \\ & + x^2 y^2 y^2 + 6 x^2 y y l^3 - 4 x^2 y y l^2 y^2 + 4 x^2 y l^4 + 2 x y^4 y l + y^4 x + 5 x y^3 y l \\ & - 3 x y^3 y^2 + 11 x y^2 y l^2 - 5 x y^2 y l y^2 + x y^2 y^2 + 10 x y y l^3 - 4 x y y l^2 y^2 \\ & + 4 x y l^4 - y^5 + y^4 + 3 y^3 y l - y^3 y^2 + 4 y^2 y l^2 - y^2 y l y^2 + 2 y y l^3) \end{aligned} \quad (1.3.9)$$

$$\begin{aligned} & \text{rho} := (\text{collect}(\text{op}(3, (1.3.9)), [y, y l, y^2], \text{distributed})) \\ \rho := & (4 x^6 + 12 x^5 + 12 x^4 + 4 x^3) y^6 + (x^4 + 6 x^3 + 5 x^2 - 1) y^5 + (6 x^4 + 16 x^3 \\ & + 12 x^2 + 2 x) y^4 y l + (-4 x^4 - 8 x^3 - 4 x^2) y^4 y^2 + (1 + x) y^4 + (8 x^4 + 16 x^3 \\ & + 8 x^2) y^3 y l^2 + (x^2 + 5 x + 3) y^3 y l + (-x^2 - 3 x - 1) y^3 y^2 + (4 x^2 + 11 x \\ & + 4) y^2 y l^2 + (-3 x^2 - 5 x - 1) y^2 y l y^2 + (x^2 + x) y^2 y^2 + (6 x^2 + 10 x \\ & + 2) y y l^3 + (-4 x^2 - 4 x) y y l^2 y^2 + (4 x^2 + 4 x) y l^4 \end{aligned} \quad (1.3.10)$$

$$\begin{aligned} & \text{ADE} := \text{subs}([y=y(x), y l=\text{diff}(y(x), x), y^2=\text{diff}(y(x), x, x)], \text{rho}) \\ \text{ADE} := & (4 x^6 + 12 x^5 + 12 x^4 + 4 x^3) y(x)^6 + (x^4 + 6 x^3 + 5 x^2 - 1) y(x)^5 + (6 x^4 \\ & + 16 x^3 + 12 x^2 + 2 x) y(x)^4 \left(\frac{d}{dx} y(x) \right) + (-4 x^4 - 8 x^3 - 4 x^2) y(x)^4 \left(\frac{d^2}{dx^2} \right. \\ & \left. y(x) \right) + (1 + x) y(x)^4 + (8 x^4 + 16 x^3 + 8 x^2) y(x)^3 \left(\frac{d}{dx} y(x) \right)^2 + (x^2 + 5 x \\ & + 3) y(x)^3 \left(\frac{d}{dx} y(x) \right) + (-x^2 - 3 x - 1) y(x)^3 \left(\frac{d^2}{dx^2} y(x) \right) + (4 x^2 + 11 x \end{aligned} \quad (1.3.11)$$

$$\begin{aligned}
& + 4) y(x)^2 \left(\frac{d}{dx} y(x) \right)^2 + (-3x^2 - 5x - 1) y(x)^2 \left(\frac{d}{dx} y(x) \right) \left(\frac{d^2}{dx^2} y(x) \right) \\
& + (x^2 + x) y(x)^2 \left(\frac{d^2}{dx^2} y(x) \right)^2 + (6x^2 + 10x + 2) y(x) \left(\frac{d}{dx} y(x) \right)^3 + (\\
& -4x^2 - 4x) y(x) \left(\frac{d}{dx} y(x) \right)^2 \left(\frac{d^2}{dx^2} y(x) \right) + (4x^2 + 4x) \left(\frac{d}{dx} y(x) \right)^4
\end{aligned}$$

NLDE finds the same ADE from p and q.

$$\begin{aligned}
& \text{CPUTime} \left(\text{NLDE:-arithmeticDalg} \left([p=0, q=0], [y(x), z(x)], w = \frac{y}{z} \right) \right) \\
& 2.266, (4x^6 + 12x^5 + 12x^4 + 4x^3) w(x)^6 + (x^4 + 6x^3 + 5x^2 - 1) w(x)^5 + (6x^4 \\
& + 16x^3 + 12x^2 + 2x) w(x)^4 \left(\frac{d}{dx} w(x) \right) + (-4x^4 - 8x^3 - 4x^2) \left(\frac{d^2}{dx^2} \right. \\
& w(x) \left. \right) w(x)^4 + (x + 1) w(x)^4 + (8x^4 + 16x^3 + 8x^2) w(x)^3 \left(\frac{d}{dx} w(x) \right)^2 \\
& + (x^2 + 5x + 3) w(x)^3 \left(\frac{d}{dx} w(x) \right) + (-x^2 - 3x - 1) w(x)^3 \left(\frac{d^2}{dx^2} w(x) \right) \\
& + (4x^2 + 11x + 4) w(x)^2 \left(\frac{d}{dx} w(x) \right)^2 + (-3x^2 - 5x - 1) w(x)^2 \left(\frac{d}{dx} \right. \\
& w(x) \left. \right) \left(\frac{d^2}{dx^2} w(x) \right) + (x^2 + x) w(x)^2 \left(\frac{d^2}{dx^2} w(x) \right)^2 + (6x^2 + 10x \\
& + 2) w(x) \left(\frac{d}{dx} w(x) \right)^3 + (-4x^2 - 4x) w(x) \left(\frac{d}{dx} w(x) \right)^2 \left(\frac{d^2}{dx^2} w(x) \right) \\
& + (4x^2 + 4x) \left(\frac{d}{dx} w(x) \right)^4 = 0
\end{aligned} \tag{1.3.12}$$

>

>

Example 4: ADE for $e^{e^x} - e$

>

> $f := \exp(\exp(\exp(x)) - \exp(1)) :$

> $L := \text{PolynomialTools:-CoefficientList}(\text{convert}(\text{series}(f, x, 80), \text{polynom}), x) :$

> $\text{numelems}(L)$

80

(1.4.1)

> $L[1..5]$

$$\left[1, e, e + \frac{(e)^2}{2}, \frac{5e}{6} + (e)^2 + \frac{(e)^3}{6}, \frac{5e}{8} + \frac{4(e)^2}{3} + \frac{(e)^3}{2} + \frac{(e)^4}{24} \right]$$

(1.4.2)

```
> CPUTime(DalgFunGuess(L, degADE=4, degPoly=0, startfromord=4, linsolver
=HardSystem))
```

$$12972.390, _C \left(\frac{d^2}{dx^2} y(x) \right) \left(\frac{d}{dx} y(x) \right)^2 y(x) - _C \left(\frac{d^3}{dx^3} y(x) \right) \left(\frac{d}{dx} y(x) \right) y(x)^2 \quad (1.4.3)$$

$$+ _C \left(\frac{d^2}{dx^2} y(x) \right) \left(\frac{d}{dx} y(x) \right) y(x)^2 - _C \left(\frac{d}{dx} y(x) \right)^4 - _C \left(\frac{d}{dx} y(x) \right)^2 y(x)^2 - _C \left(\frac{d}{dx} y(x) \right)^3 y(x) + _C \left(\frac{d^2}{dx^2} y(x) \right)^2 y(x)^2 = 0$$

```
> CPUTime(DalgFunGuess(L[1..20], degADE=4, degPoly=0, startfromord=4,
allPolyDeg=true, sparsity=3/4, maxIteration=10))
256.531, FAIL
```

(1.4.4)

▼ modDalgFunGuess

```
[>
```

▼ Example 5: OEIS [A189281](#)

```
[>
> T := readdata("b189281.txt", integer, 2) :
> L := map(t->t[2], T) :
> numelems(L)
301
```

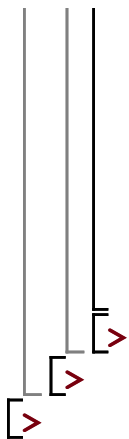
(2.1.1)

```
> L[-1] #the last term in the list
11372069544670615182450141371651349899813311450244396383365131233787919\ (2.1.2)
03963787297097950914614622744980916245525095837059122625005087256009\
61267583819951354960326626583396441011329385038320608019808450146379\
81363927893371454163853400798548859033415579591422667854989517752461\
11392830369578261524732922126589677966923150803305798073113073143330\
01266371294938427626025774479608026492895343733562842729487316487770\
77733286230349235354762502184041098538039037527828777121056595891273\
03659194955994905489080540135700897583959649922532184971527954332600\
81844544920264233182418342148466552185999146775891138264272876438194
```

```
> CPUTime(modDalgFunGuess(L, degADE=1, degPoly=11, startfromord=19, modulus
=569))
```

(2.1.3)

$$\begin{aligned}
& 0.907, (437x^2 + 526x + 303) \left(\frac{d^2}{dx^2} y(x) \right) + (46x^{11} + 194x^{10} + 522x^9 + 289x^8 \\
& + 231x^7 + 62x^6 + 291x^5 + 336x^4 + 4x^2) \left(\frac{d^{17}}{dx^{17}} y(x) \right) + (287x^{11} + 102x^{10} \\
& + 111x^9 + 434x^8 + 137x^7 + 3x^6 + 4x^5) \left(\frac{d^{18}}{dx^{18}} y(x) \right) + (568x^{11} + 568x^{10} \\
& + x^9 + x^8) \left(\frac{d^{19}}{dx^{19}} y(x) \right) + (51x^{11} + 159x^{10} + 534x^9 + 360x^8 + 546x^7 + 20x^6 \\
& + 236x^5 + 343x^4 + 359x^3 + 60x^2 + 383x + 381) \left(\frac{d^{15}}{dx^{15}} y(x) \right) + (16x^{11} \\
& + 25x^{10} + 12x^9 + 340x^8 + 254x^7 + 238x^6 + 193x^5 + 200x^4 + 378x^3 + 563x^2 \\
& + 128x + 565) \left(\frac{d^{16}}{dx^{16}} y(x) \right) + (152x^{11} + 194x^{10} + 506x^9 + 175x^8 + 534x^7 \\
& + 373x^6 + 374x^5 + 143x^4 + 530x^3 + 339x^2 + 415x + 248) \left(\frac{d^{12}}{dx^{12}} y(x) \right) \\
& + (232x^{11} + 404x^{10} + 313x^9 + 245x^8 + 96x^7 + 128x^6 + 515x^5 + 380x^4 \\
& + 212x^3 + 174x^2 + 188x + 479) \left(\frac{d^{13}}{dx^{13}} y(x) \right) + (56x^{11} + 215x^{10} + 190x^9 \\
& + 200x^8 + 263x^7 + 533x^6 + 327x^5 + 262x^4 + 47x^3 + 347x^2 + 461x + 501) \\
& \left(\frac{d^{14}}{dx^{14}} y(x) \right) + (47x^{10} + 484x^9 + 548x^8 + 433x^7 + 511x^6 + 47x^5 + 306x^4 \\
& + 196x^3 + 4x + 336) \left(\frac{d^{10}}{dx^{10}} y(x) \right) + (122x^{11} + 111x^{10} + 512x^9 + 110x^8 \\
& + 249x^7 + 206x^6 + 90x^5 + 118x^4 + 124x^3 + 207x^2 + 508x + 24) \left(\frac{d^{11}}{dx^{11}} \right. \\
& \left. y(x) \right) + (374x^7 + 45x^6 + 275x^5 + 500x^4 + 110x^3 + 390x^2 + 513x \\
& + 374) \left(\frac{d^7}{dx^7} y(x) \right) + (157x^8 + 445x^7 + 229x^6 + 105x^5 + 332x^4 + 38x^3 \\
& + 218x^2 + 286x + 55) \left(\frac{d^8}{dx^8} y(x) \right) + (307x^9 + 34x^8 + 489x^7 + 336x^6 \\
& + 321x^5 + 4x^4 + 216x^3 + 189x^2 + 511x + 143) \left(\frac{d^9}{dx^9} y(x) \right) + (302x^3 \\
& + 507x^2 + 260x + 519) \left(\frac{d^3}{dx^3} y(x) \right) + (431x^4 + 366x^3 + 25x^2 + 240x
\end{aligned} \tag{2.1.3}$$



$$\begin{aligned} &+ 133) \left(\frac{d^4}{dx^4} y(x) \right) + (451 x^5 + 388 x^4 + 506 x^3 + 281 x^2 + 188 x \\ &+ 125) \left(\frac{d^5}{dx^5} y(x) \right) + (14 x^6 + 235 x^5 + 94 x^4 + 477 x^3 + 554 x^2 + 532 x \\ &+ 358) \left(\frac{d^6}{dx^6} y(x) \right) = 0 \end{aligned}$$